Optimized Translation of MATLAB to RISC-V Architecture for Brain-Computer Interfaces: Final Report

Kenan Erol
kenan.erol@yale.edu

Advisor: Rajit Manohar
rajit.manohar@yale.edu

A Senior Thesis as a partial fulfillment of requirements for the Bachelor of Science in Computer Science

Department of Computer Science
Yale University
April 30, 2026

Acknowledgements

I want to express my gratitude to my family for their continuous support and guidance. Special thanks to the Yale Department of Computer Science and the Yale Department of Electrical and Computer Engineering for providing a fantastic environment for academic growth. I'm grateful for the opportunities, scholarships, and resources that have enriched my time here. I would especially like to thank my advisor, Professor Rajit Manohar, for his guidance during my time in the major and throughout this project.

Computational resources were provided by the Yale Grace cluster. This project was undertaken as part of the CPSC 4900 Senior Thesis course at Yale University under the supervision of the Director of Undergraduate Studies, Professor Theodore Kim.

As approved by Professor Manohar, Claude Opus 4.6, an AI coding assistant, was used as an interactive debugging and pair programming tool to accelerate the process of identifying syntax issues and parse error logs. AI was also used to accelerate understanding of RVV intrinsics. Actual code changes and vectorization were manually confirmed to ensure integrity of results.

Contents

1 Introduction **5**

2 Background **7**

3 Design **9**

4 Results **11**
- 4.1 Additional Vectorization 11
- 4.2 Total Runtime Comparison 11
  - 4.2.1 The `divide` Anomaly 13
- 4.3 Vectorization Effectiveness 13
  - 4.3.1 Ratio Shifts Due to Compiler Improvements 15
  - 4.3.2 The `isEqual` Anomaly 16
- 4.4 Toolchain-Induced Vectorization Regressions 17
- 4.5 Absolute Cycle Counts 17
- 4.6 TS-traversal Compiler 18
  - 4.6.1 C-Octave Suite Residuals 21

5 Related Work **22**
- 5.1 Translation for High-Level Languages 22

6 Conclusions **24**
- 6.1 Future Work 24

A Full Binary Benchmark Data **28**
- A.1 Full Binary Ratio Comparison 28
- A.2 Binary Operation Cycle Counts 33

Optimized Translation of MATLAB to RISC-V Architecture for Brain-Computer Interfaces

Kenan Erol

Abstract

Brain computer interfaces (BCIs) offer significant clinical and therapeutic potential for neurological disorders such as epilepsy and stroke. Fully implantable BCIs impose severe limitations on power dissipation to prevent irreversible thermal damage to surrounding brain tissue. As a result, BCI hardware needs to balance real-time signal processing with ultra-low power consumption. The Hardware Architecture for LOw Power BCIs (HALO) project addresses these constraints through a design that combines application-specific hardware accelerators with a low-power RISC-V microcontroller equipped with the RISC-V Vector extension. Despite the powerful capabilities of the HALO architecture, there is a significant toolchain gap between the computational environments preferred by neuroscientists and the low-level constraints of the HALO hardware that proves a hindrance to the widespread adoption of the technology. Algorithm development typically occurs in dynamically typed languages such as MATLAB, whereas HALO hardware requires statically typed, optimized, and vectorized C code. This report details the development of an end-to-end compilation pipeline engineered to bridge this divide. The pipeline uses a TypeScript-based compiler and an RVV-optimized C matrix library to translate MATLAB algorithms into power-efficient RISC-V binaries. The report discusses the architectural design of the pipeline, benchmarks on the Spike instruction-set architecture simulator, investigates execution-cycle metrics, discusses translation benchmarks, and presents next steps for the research.

1 Introduction

Recent work in computational neuroscience and embedded systems has enabled the development of BCIs capable of real-time communication between biological neural networks and digital processing units. BCIs are central to closed-loop therapeutic applications, where the device must continuously process high-density, multi-channel neural recordings, execute complex feature extraction algorithms, and determine appropriate electrical stimulation parameters to mitigate neurological events such as epileptic seizures [1]. Fully implantable BCIs must meet physical and thermal constraints to ensure brain tissue safety, reduce surgical complexity, and extend device longevity. The thermal dissipation limit is the most critical constraint. To prevent irreversible necrosis of the surrounding brain tissue, the surface temperature of the implant must not exceed 1 °C, thereby strictly capping the total device power budget at approximately 15 mW [2].

The HALO BCI project was engineered specifically to satisfy these conflicting demands. The HALO architecture differs from traditional application-specific integrated circuit (ASIC) designs, which often lack the flexibility to adapt to new algorithms. Instead, HALO uses a modular array of hardware processing elements (PEs) controlled by a low-power RISC-V microcontroller [1]. The microcontroller uses the RISC-V Vector (RVV) extension to execute data-parallel signal-processing workloads, which minimizes instruction-fetch overhead and enables rapid transitions into low-power states whenever possible.

However, the clinical and research efficiency of HALO is currently bottlenecked by a toolchain disparity. Neuroscientists use languages like MATLAB for their intuitive syntax, signal-processing toolboxes, and dynamic typing. In contrast, HALO needs statically typed C code that contains RVV intrinsics. The manual translation of signal processing algorithms, such as Fast Fourier transforms and matrix operations, from MATLAB to vectorized C is an error-prone, labor-intensive process that impedes the rapid deployment of new clinical approaches.

This research addresses the toolchain gap by engineering an automated compilation pipeline that translates standard MATLAB scripts into RVV-accelerated RISC-V binaries for HALO. The proposed solution relies on the integration and expansion of two software modules previously developed in isolation: a compiler capable of parsing MATLAB syntax into equivalent C code [3], and a computational C matrix library that maps standard linear algebra operations directly to RVV assembly instructions [4].

Several milestones were completed during the course of this project. The matrix library has been successfully replicated, integrated into a benchmarking environment that uses Spike, the official RISC-V instruction set architecture (ISA) simulator, and been vectorized further. Performance evaluations reveal substantial reductions in execution cycle counts. In addition, debugging efforts have identified and fixed algorithmic limitations in

the translation pipeline after a full correctness and speed benchmark was obtained. This final report establishes the foundational background, outlines the pipeline's structural design, analyzes current benchmarking results, and discusses future development plans.

2 Background

To understand the design constraints and operational mechanics of the proposed translation pipeline, we first review the architectural constraints of the target BCI hardware, the computational paradigms introduced by RVV, and the foundational principles of the compilers used in this research.

The fundamental engineering challenges in the development of implantable BCIs is the strict management of thermal dissipation. The human cerebral cortex is highly sensitive to temperature fluctuation, and prolonged exposure to high temperatures can cause tissue damage. Consequently, international medical standards dictate that an implant must not raise the local tissue temperature by more than one degree Celsius, effectively restricting the total power consumption of the entire BCI system—including amplifiers, analog-to-digital converters (ADCs), wireless telemetry, and digital signal processing (DSP) units—to roughly fifteen milliwatts [2].

Historically, BCI designers have relied on custom-designed ASICs to meet these power requirements [5] [6] [7]. While ASICs offer high energy efficiency, they inherently lack flexibility. If a neuroscientist develops a superior seizure-prediction algorithm or wishes to repurpose the implant for a different disorder, a completely new physical chip must be engineered, fabricated, and implanted [1]. General purpose microprocessors offer the required flexibility through software updates but consume power orders of magnitude beyond the safe physiological limit [2].

HALO resolves this through aggressive hardware-software co-design. The architecture is built around a configurable interconnect that links a heterogeneous array of application specific PEs [1]. These PEs are optimized for the specific computational kernels most frequently encountered in neural signal processing, such as FFT, XCOR, and Butterworth bandpass filtering [2]. These accelerators can be dynamically chained together to form highly efficient data flow pipelines tailored to specific clinical tasks [1].

To manage these pipelines and execute computational tasks that fall outside the capabilities of a standalone PE, HALO uses a low power RISC-V microcontroller [1]. To ensure that this software-driven execution does not violate the power budget, the microcontroller relies heavily on RVV. Unlike traditional single instruction, multiple data (SIMD) architectures, which utilize fixed width vector registers and require code rewrites when the vector width changes for different hardware generations, RVV introduces a vector-length agnostic programming model [8, 9].

In the context of BCI workloads, which involve matrix multiplications and convolutions over various window sizes, RVV allows the processor to load large, contiguous segments of multi-channel neural data into vector registers and apply a single arithmetic instruction across the entire dataset simultaneously. This significantly reduces the dynamic power consumed by the processor's instruction fetch and decode units. By replac-

ing hundreds or thousands of scalar loop iterations with a single vector instruction, the processor minimizes the total number of active clock cycles required to complete a given task. Consequently, computational algorithms execute much faster, allowing the microcontroller to aggressively switch to low-power sleep states between neural data sampling intervals, which is crucial for extending the lifespan of the implantable battery.

Despite these hardware features, the platform remains inaccessible to neuroscientists. Clinical algorithms are developed using high-level, dynamically typed languages like MATLAB, which abstract away memory management and hardware-specific optimizations that are crucial for running the algorithms on silicon. The automated pipeline proposed in this research bridges this gap by integrating two existing software repositories. The first is ts-traversal, a TypeScript-based source-to-source compiler that parses MATLAB syntax, infers static types, and generates functional C code [3]. The second is the RISCV-Matrix C library, which maps standard mathematical operations directly to RVV assembly instructions [4]. Critical fixes have been made in both the translation logic and the vectorization coverage.

3 Design

The objective of this research is to design the architecture of and to implement a cohesive, automated compilation pipeline capable of translating high-level algorithms into low-level optimized silicon instructions. This section details the system architecture, focusing on the transformation mechanisms and the underlying platform dependencies.

The compiler pipeline is implemented as a continuous, directed data flow that steps from the original MATLAB file into the RISC-V binary. The system architecture can be visualized through a structured sequence of state transitions and module interactions in Figure 3.1. The pipeline starts by taking in raw .m files authored by clinical researchers. These files contain standard MATLAB syntax, incorporating dynamic array allocations, calls to proprietary signal processing toolboxes and matrix arithmetic.

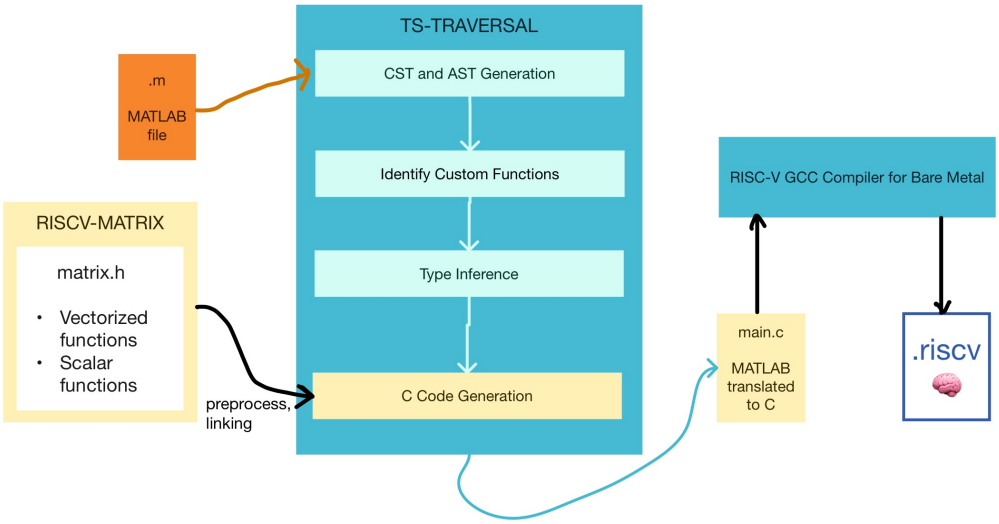

Figure 3.1: System architecture of the MATLAB-to-RISC-V compilation pipeline. Raw `.m` files are parsed and type-inferred by the ts-traversal engine. Afterwards, the MATLAB operations are mapped to RVV intrinsics via the RISCV-Matrix library. The resulting main.c file is compiled by the RISC-V GCC bare metal toolchain into a single binary, validated on the Spike ISA simulator.

These raw files are passed into ts-traversal [3], which is the first component of the translation pipeline. ts-traversal executes a multi-phase transformation. First, it uses the

TreeSitter and TreeSitter-Matlab parsing libraries to create a concrete syntax tree (CST). This representation captures the complete syntactic structure of the input MATLAB file and the algorithms/functions it contains. Afterwards, it recursively identifies and extracts the signatures of custom user-defined `.m` functions, which enables the later stage of type inference to propagate data types across nested function calls throughout the CST. Then, the engine performs type inference on the dynamically typed input MATLAB code. The inference algorithms in ts-traversal traverse the CST to understand the maximal dimensions, memory allocation requirements, and fundamental data types of all MATLAB variables, mapping them to statically typed C constructs. The output of this phase is standard C code. The semantic intent of the MATLAB script is entirely preserved and the available vectorized and scalar functions are mapped via RISCV-Matrix library's `matrix.h` header, which contains highly optimized, inline RVV C intrinsics as well as reference scalar implementations of functions. This transformation effectively converts the MATLAB into heavily accelerated C code explicitly built for the parallel execution capabilities of HALO's microcontroller.

The final phase involves compiling and deploying the codebase. The generated main.c file is compiled via the RISC-V GNU Compiler toolchain. The compiler statically links the optimized vector library to produce a highly efficient RISC-V binary. Thus, the pipeline generates an executable containing the exact algorithms authored by neuroscientists in Matlab while meeting the power and latency requirements of the implanted chip.

The validation, execution modeling, and accurate instruction-level benchmarking of the generated binaries are conducted using the Spike ISA simulator, since the actual HALO silicon is not readily available. Spike provides a fully functional, instruction-accurate model of a RISC-V core with the RVV extension, enabling precise tracking of executed instructions, vector register utilization, and overall execution cycles. This enables the continuous, iterative refinement of both the source-to-source compiler and the vectorization library without requiring constant access to the physical test hardware.

4 Results

The first critical phase of this project was to replicate the original RISCV-Matrix execution environment and rebuild the benchmark suite under the current toolchain. The old speed and accuracy results of RISCV-Matrix were generated using an older cross-compiler toolchain and test suite configuration. This section compares those original results, designated the Grace baseline (commit `113f4c66`, December 2022), against the current, optimized pipeline (April 2026). All cycle counts were measured on the Spike RISC-V ISA simulator with a vector length of VLEN=1024.

Next, the focus was to benchmark the ts-traversal translation pipeline with various MATLAB and Octave tests for correctness and speed. Finally, some non-vectorized functions from RISCV-Matrix were vectorized and mapped onto the ts-traversal pipeline.

4.1 Additional Vectorization

The Grace baseline shipped vectorized kernels for the full binary, arithmetic, and comparison set. The work reported here extends the vectorized set by adding `_vec` entry points for `transposeM`, `ctransposeM`, `meanM`, `varM`, `minM`, `maxM`, `minV`, `maxV`, `stdM`, and `popstdM`. These additions now appear as new rows in Tables 4.2 and 4.6.

The ten new entry points land in April 2026. Accuracy tests pass for all of them at every size from $N = 1$ to $N = 1{,}050{,}625$ except for two tests at 127x127 matrices: `varM`, `stdM`, and `mtimesM` (a vectorized kernel written before this research) have their vec and ref outputs off by a factor of at most 8.941×10^{-8}, 2.444×10^{-9} and 1.714×10^{-7} respectively.

The ten new functions have not all been vectorized for all possible dimensions and types. For `transposeM` and `ctransposeM`, all 2D types have been vectorized while all 1D inputs fall back to the reference implementation. For `meanM` and `varM`, 1D doubles and all 2D types have been vectorized. For `minM/V` and `maxM/V`, all but 1D and 2D complex inputs have been vectorized due to the need for `carg` for complex comparison. `stdM` and `popstdM` inherit the exact same constraints as `varM`, remaining fully vectorized only for 1D doubles and all 2D matrix types.

4.2 Total Runtime Comparison

Table 4.1 details the total runtime comparison, displaying the specific cycle counts and the corresponding percentage changes in execution speed.

Table 4.1: Total program runtime comparison: Grace (Dec 2022) vs Current (April 2026). Cycle counts measured on Spike simulator with VLEN=1024.

Test	Grace (Gcycles)	Current (Gcycles)	Δ (Gcycles)	Change (%)
abs	9.86	9.25	-0.61	-6.2% ↓
and	2.22	1.91	-0.31	-14.2% ↓
arg	6.48	5.33	-1.14	-17.7% ↓
ceil	4.96	4.14	-0.82	-16.5% ↓
ctranspose	—	3.56	—	—
divide	5.47	5.64	+0.16	+3.0% ↑
equal	4.75	4.31	-0.44	-9.3% ↓
floor	4.97	4.15	-0.81	-16.4% ↓
full_duplicate	4.32	3.48	-0.84	-19.4% ↓
isEqual	3.79	3.14	-0.65	-17.2% ↓
matlab_ge	4.37	3.65	-0.72	-16.6% ↓
matlab_gt	4.36	3.63	-0.72	-16.6% ↓
matlab_le	4.36	3.64	-0.72	-16.5% ↓
matlab_lt	4.34	3.63	-0.72	-16.5% ↓
maxV	—	26.90	—	—
max	—	26.93	—	—
mean	—	3.50	—	—
minV	—	27.52	—	—
min	—	27.12	—	—
minus	6.18	5.11	-1.07	-17.3% ↓
mtimes	3.07	3.01	-0.07	-2.3% ↓
ne	4.81	4.36	-0.44	-9.2% ↓
not	1.51	1.28	-0.23	-15.2% ↓
octave_ge	51.90	51.60	-0.30	-0.6% ≈
octave_gt	51.90	51.60	-0.30	-0.6% ≈
octave_le	51.94	51.67	-0.27	-0.5% ≈
octave_lt	51.94	51.67	-0.27	-0.5% ≈
ones	0.05	0.03	-0.02	-44.3% ↓
or	2.15	1.93	-0.22	-10.1% ↓
pairwise_max	54.36	53.60	-0.76	-1.4% ↓
pairwise_min	54.42	53.67	-0.74	-1.4% ↓
plus	6.19	5.12	-1.07	-17.3% ↓
popstd	—	4.02	—	—
round	4.95	4.05	-0.90	-18.2% ↓
scale	1.98	1.76	-0.22	-10.9% ↓
std	—	3.69	—	—
times	6.37	5.33	-1.05	-16.4% ↓
transpose	—	3.55	—	—
true	1.54	1.35	-0.19	-12.4% ↓
var	—	3.69	—	—
Comparable Total	**419.50**	**403.01**	**-16.49**	**-3.9%**
New tests only	—	*130.49*	—	—

An analysis of Table 4.1 reveals several insights about the current state of the compilation pipeline. Across the entire test suite, the overall execution time decreased by approximately 3.9 percent, dropping from 419.50 Giga-cycles to 403.01 Giga-cycles. RISC-V GNU compiler collection (GCC) updates played a crucial part in these improvements. The distribution of these performance gains is non-uniform. For instance, the majority of the standard arithmetic and logical operations experienced moderate speedups ranging from 10-20 percent. All four core arithmetic tests dropped by 10-17%, consistent with the GCC toolchain improvements. This demonstrates that the GCC updates benefit both vectorized and scalar code paths across these operations.

The four comparison tests utilizing Octave-mode semantics–`octave_ge`, `octave_gt`, `octave_le`, and `octave_lt`–showed negligible improvement ($< 1\%$). These tests dominate the total runtime across the test suite at approximately 51 Giga-cycles each (versus $\sim$3.6 Giga-cycles for their MATLAB-mode counterparts) because Octave comparison semantics for complex numbers require computing the magnitude ($|z| = \sqrt{\mathrm{re}^2 + \mathrm{im}^2}$) for both operands and, when magnitudes are equal, resolving ties via the phase angle ($\mathrm{atan2}(\mathrm{im}, \mathrm{re})$) [10]. The phase-angle path (`arg`) required by Octave currently executes as a scalar libc arg call because it was never mapped to RVV intrinsics, making it the dominant cost. In contrast, MATLAB-mode comparisons only compare the real parts of complex numbers, requiring a single vector comparison instruction.

4.2.1 The `divide` Anomaly

The `divide` test is the only operation that exhibited a total runtime increase ($+3.0\%$, from 5.47 to 5.64 Gcycles). This regression is localized to complex division reference counts – the reference cycle counts for complex-involving type combinations approximately doubled between Grace and current (e.g., `complex`×`complex ref`: 36.1M $\rightarrow$ 61.3M cycles), while the vectorized cycle counts remained stable ($\sim$8M cycles). This suggests that the updated GCC generates less efficient scalar code for complex division.

In contrast, the `int`×`int` reference for `divide` dropped from 14.4M to 4.8M (-67%), indicating that the compiler dramatically improved integer-only scalar division. This shows that the same toolchain update can simultaneously upgrade and degrade performance across different data type combinations, even within a single function.

4.3 Vectorization Effectiveness

Further analysis was conducted to evaluate the vectorization implementation relative to scalar execution across different variable types. Tables 4.2 and 4.3 present the vector-to-reference execution ratios at the largest tested matrix size (1025×1025). A ratio less than 1.0 indicates that the vectorized implementation is faster than the scalar reference. The tables compare Grace and current ratios side-by-side.

Note on type combinations: for binary operations (e.g., `plusM`, `divideM`), the test suite evaluates every combination of input types. The notation "int $\times$ double" means the left operand matrix contains integers and the right operand matrix contains doubles. The result type is determined by C type promotion rules.

Table 4.2: Vec/Ref ratios at $N \approx 1{,}050{,}625$ for unary operations. Ratio < 1 means vectorized is faster. Δ shows change from Grace to Current.

	int			double			complex		
Test	Grace	Cur.	Δ	Grace	Cur.	Δ	Grace	Cur.	Δ
abs	0.1932	0.1908	-0.0023	0.3897	0.3868	-0.0029	0.0044	0.0042	-0.0002
arg	0.0625	0.0732	+0.0107	0.0631	0.0740	+0.0109	1.0000	1.0000	+0.0000
ceil	0.6254	0.9103	+0.2849	0.0959	0.0942	-0.0017	0.0971	0.0939	-0.0032
ctranspose	—	0.2773	—	—	0.4356	—	—	0.4949	—
floor	0.6254	0.9103	+0.2850	0.0959	0.0942	-0.0017	0.0959	0.0922	-0.0037
full_duplicate	0.6254	0.9103	+0.2850	0.6253	0.9103	+0.2850	0.6253	0.6054	-0.0200
maxV	—	0.9011	—	—	0.9732	—	—	1.0000	—
max	—	0.0447	—	—	0.0793	—	—	1.0000	—
mean	—	0.0447	—	—	0.0830	—	—	0.1644	—
minV	—	0.9011	—	—	0.9732	—	—	1.0000	—
min	—	0.0447	—	—	0.0793	—	—	1.0000	—
not	0.3550	0.3523	-0.0027	0.3024	0.8529	+0.5505	0.2994	0.8406	+0.5412
ones	0.3873	0.9225	+0.5352	—	—	—	—	—	—
popstd	—	0.9902	—	—	0.9900	—	—	0.9880	—
round	0.6254	0.9103	+0.2850	0.1057	0.1148	+0.0091	0.0992	0.1070	+0.0078
std	—	0.0758	—	—	0.1371	—	—	0.1079	—
transpose	—	0.2773	—	—	0.4356	—	—	0.6141	—
true	0.3411	0.3385	-0.0026	0.3130	0.8529	+0.5399	0.2899	0.8406	+0.5507
var	—	0.0765	—	—	0.1384	—	—	0.1092	—

Table 4.3: Vec/Ref ratios at $N \approx 1{,}050{,}625$ for binary operations (diagonal type combos). Ratio < 1 means vectorized is faster.

	int × int			dbl × dbl			cpx × cpx		
Test	Grace	Cur.	Δ	Grace	Cur.	Δ	Grace	Cur.	Δ
and	0.6375	0.6314	-0.0061	0.4931	2.6775	+2.1844	0.3653	0.4173	+0.0519
divide	0.2802	0.8232	+0.5430	0.3249	0.3218	-0.0031	0.2351	0.1332	-0.1019
equal	0.1729	0.1384	-0.0345	0.1913	0.1502	-0.0411	0.1606	0.1289	-0.0317
isEqual	0.6711	0.6575	-0.0135	0.7971	0.7313	-0.0658	1.1719	1.1207	-0.0512
matlab_ge	0.2817	0.2792	-0.0025	0.2533	0.8275	+0.5742	0.2539	0.2510	-0.0029
matlab_gt	0.2817	0.2792	-0.0025	0.2533	0.8275	+0.5742	0.2539	0.2510	-0.0029
matlab_le	0.2817	0.2792	-0.0025	0.2767	0.8275	+0.5508	0.2774	0.2742	-0.0032
matlab_lt	0.2817	0.2792	-0.0025	0.2767	0.8275	+0.5508	0.2774	0.2742	-0.0032
minus	0.7619	0.7580	-0.0039	0.3249	0.3218	-0.0031	0.3921	0.4898	+0.0977
mtimes	0.0594	0.0481	-0.0113	0.0521	0.0401	-0.0120	0.0443	0.0318	-0.0126
ne	0.1632	0.1321	-0.0311	0.1799	0.1429	-0.0370	0.1529	0.1237	-0.0292
octave_ge	0.2817	0.2792	-0.0025	0.2533	0.8275	+0.5742	0.1416	0.1230	-0.0186
octave_gt	0.2817	0.2792	-0.0025	0.2533	0.8275	+0.5742	0.1416	0.1230	-0.0186
octave_le	0.2817	0.2792	-0.0025	0.2767	0.8275	+0.5508	0.1410	0.1222	-0.0188
octave_lt	0.2817	0.2792	-0.0025	0.2767	0.8275	+0.5508	0.1410	0.1222	-0.0188
or	0.6478	0.6416	-0.0062	0.5182	0.5107	-0.0075	0.4620	0.4475	-0.0145
pairwise_max	0.0011	0.0008	-0.0003	0.0040	0.0041	+0.0000	0.1347	0.1166	-0.0182
pairwise_min	0.0011	0.0008	-0.0003	0.0040	0.0041	+0.0000	0.1341	0.1158	-0.0183
plus	0.7619	0.7580	-0.0039	0.3249	0.3218	-0.0031	0.3921	0.4898	+0.0977
scale	0.8068	0.8039	-0.0029	0.3532	0.3505	-0.0027	0.3102	0.3072	-0.0030
times	0.7619	0.7580	-0.0039	0.3249	0.3218	-0.0031	0.3017	0.2973	-0.0044

A comprehensive breakdown of all nine type combinations for each binary operation is provided in Table A.1 in the Appendix.

4.3.1 Ratio Shifts Due to Compiler Improvements

While the vectorized implementations maintained approximately constant cycle counts between Grace and current (for the previously existing vectorized workloads), several vec/ref ratios increased significantly–meaning the scalar *reference* code got faster, narrowing the vectorization advantage. Three distinct patterns were identified:

Pattern 1: Integer memory operations. The integer reference cycle counts dropped uniformly for the `ceil`, `floor`, `round`, `full_duplicate`, and `ones` tests. Many of these tests saw their ratios jump by $\sim 29\%$ across int and double types. This is likely because GCC optimized memory load/store type operations since the Grace era.

Pattern 2: Logical operations on non-integer types. A similar ref speedup was observed for logical operations on double and complex types. The cycle count for `not` and `true` decreased by ∼65% (e.g., `not` double ref: 10.3M → 3.6M) while the vectorized counts stayed at ∼3.1M. This caused the double/complex ratios to jump from ∼0.30 to ∼0.85. The GCC compiler now seems to optimize these operations more aggressively.

Pattern 3: Binary operations. Several binary operations exhibited compiler-driven ref improvements. `matlab_ge` through `matlab_lt` (dbl×dbl), and `octave_ge` through `octave_lt` (dbl×dbl) all showed reference speedups of 3–5×, pushing their ratios from ∼0.25 to ∼0.83. The full details on binary operation cycles and ratios are available in the Appendix (Table A.1).

In all cases, the vectorized code did not get slower–the reference code simply got faster. The absolute performance of the library improved or stayed steady, but the relative advantage of explicit vectorization narrowed for these specific combinations.

4.3.2 The `isEqual` Anomaly

The `isEqual` test consistently returns a vec/ref ratio greater than 1.0 for complex type combinations (e.g., complex×complex: 1.172 in Grace, 1.121 in current), indicating that the vectorized implementation is slower than scalar. This turns out to be caused by the test harness, not the implementation. The test harness short-circuits the comparison, so the vectorized code never gets to the expensive part of the comparison.

The evidence shows that the cycle counts for `isEqual` complex×complex remain **exactly 75 cycles (vec) and 64 cycles (ref) at every single matrix size tested**–from $N = 1$ through $N = 1{,}050{,}625$:

Table 4.4: The `isEqual` complex×complex cycle counts remain constant across all matrix sizes, proving short-circuit evaluation.

Size N	Vec (cycles)	Ref (cycles)	Ratio
1	75	64	1.172
9	75	64	1.172
16,384	75	64	1.172
1,050,625	75	64	1.172

If the implementations were actually traversing the data, the cycle counts would scale linearly with N. The lack of linear scaling shows that both vec and ref implementations exit after comparing only the *first element*. An examination of the test data generation confirmed that the `isEqual` test compares matrices populated with entirely divergent data streams, causing an immediate short-circuit mismatch.

The 17% performance gap (75 vs. 64 cycles) is attributable to *vector setup overhead*. `isEqual_vec` executes vector length configuration and stride-loading instructions before performing comparisons. However, the scalar path simply loads and compares two values, so it does not have the same overhead cost. In workloads where equality must be verified across a full traversal, the vectorized implementation vastly outperforms scalar execution, as can be seen in the complex×double combination achieving a ratio of 0.015 (66× speedup).

4.4 Toolchain-Induced Vectorization Regressions

While the preceding section described cases where the vec/ref ratio *increased but remained below 1.0* (vec still faster), three specific type combinations crossed the 1.0 threshold entirely–meaning the scalar reference now outperforms the vectorized implementation:

Table 4.5: Type combinations where the scalar reference became faster than the vectorized implementation in the current toolchain.

Test	Type Combo	Grace		Current	
		Ratio	Ref (cycles)	Ratio	Ref (cycles)
and	dbl × dbl	0.493	10.9M	2.678	2.0M
and	cpx × dbl	0.408	13.3M	2.464	2.2M
or	cpx × dbl	0.477	11.4M	2.392	2.2M

In all three cases, the vectorized cycle counts remained approximately constant (∼5.3M for `and`, ∼5.4M for `or`). The regressions are entirely driven by GCC's updated optimizer reducing the reference cycle counts by a factor of five to six. It is expected that the "scalar" reference is no longer scalar: GCC 15's RISC-V backend auto-vectorizes it, and the resulting code beats the library's hand-written kernel.

This has two implications. First, compiler evolution can erode the benefits of manual vectorization. Second, the hand-written kernels here use `LMUL=8`, which the RISCV-Matrix README warns will "cause extra loads and stores to be called implicitly" when the kernel uses more than a handful of live vector registers [4]. Investigating the LMUL choice is therefore a suggested ratio recovery path for future work.

4.5 Absolute Cycle Counts

Table 4.6 presents the absolute vectorized and reference cycle counts at the largest matrix size for both Grace and current environments to display the raw performance data.

Table 4.6: Vectorized and reference cycle counts at $N \approx 1{,}050{,}625$ for unary operations (Grace vs Current).

	int				double				complex			
Test	G.vec	C.vec	G.ref	C.ref	G.vec	C.vec	G.ref	C.ref	G.vec	C.vec	G.ref	C.ref
abs	2.0M	2.0M	10.3M	10.3M	4.0M	3.9M	10.2M	10.1M	4.0M	4.0M	913.9M	948.4M
arg	4.0M	4.0M	64.3M	54.3M	4.0M	3.9M	63.2M	53.2M	125.8M	109.5M	125.8M	109.5M
ceil	2.0M	2.0M	3.2M	2.2M	4.0M	4.0M	42.2M	42.5M	8.1M	8.0M	83.3M	85.2M
ctranspose	—	2.0M	—	7.2M	—	4.0M	—	9.1M	—	8.0M	—	16.1M
floor	2.0M	2.0M	3.2M	2.2M	4.0M	4.0M	42.2M	42.5M	8.1M	8.0M	84.4M	86.7M
full_duplicate	2.0M	2.0M	3.2M	2.2M	3.9M	3.9M	6.3M	4.3M	7.9M	7.8M	12.6M	12.9M
maxV	—	5.7M	—	6.3M	—	7.2M	—	7.4M	—	1928.8M	—	1928.8M
max	—	445.9K	—	10.0M	—	874.6K	—	11.0M	—	1933.2M	—	1933.2M
mean	—	516.7K	—	11.6M	—	871.8K	—	10.5M	—	1.7M	—	10.5M
minV	—	5.7M	—	6.3M	—	7.2M	—	7.4M	—	1979.8M	—	1979.8M
min	—	445.9K	—	10.0M	—	874.6K	—	11.0M	—	1949.2M	—	1949.2M
not	3.0M	3.0M	8.6M	8.6M	3.1M	3.1M	10.3M	3.6M	3.2M	3.2M	10.7M	3.8M
ones	2.0M	2.0M	5.1M	2.1M	—	—	—	—	—	—	—	—
popstd	—	15.2M	—	15.4M	—	14.5M	—	14.7M	—	31.2M	—	31.6M
round	2.0M	2.0M	3.2M	2.2M	4.2M	4.1M	39.8M	35.9M	8.4M	8.2M	84.9M	77.0M
std	—	1.2M	—	15.4M	—	2.0M	—	14.7M	—	3.4M	—	31.6M
transpose	—	2.0M	—	7.2M	—	4.0M	—	9.1M	—	8.0M	—	12.9M
true	3.0M	3.0M	8.9M	8.9M	3.1M	3.1M	10.0M	3.6M	3.2M	3.2M	11.0M	3.8M
var	—	1.2M	—	15.2M	—	2.0M	—	14.5M	—	3.4M	—	31.2M

These raw counts confirm that vectorized implementations maintained or slightly improved their absolute performance, while compiler improvements to the scalar reference code drove all observed ratio shifts.

4.6 TS-traversal Compiler

Beyond the resolution of benchmarking anomalies, significant debugging efforts have focused on addressing algorithmic failures within the ts-traversal compiler phase. When the project was inherited, the TypeScript traversal would emit C that compiled but failed on several fundamental constructs: higher-dimensional indexing (3-D and 4-D array subscripts), various standalone issues that propagated across multiple functions like off-by-one for loop iteration, missing column vector support, missing function mapping, and row major vs. column major representation across MATLAB and C. There were also some tests that were producing different results across the MATLAB and generated C files due to a difference in the type of BLAS (Basic Linear Algebra Subprogram) libraries used between MATLAB (OPENBLAS, requires POSIX) and our bare metal RISC-V binaries (LAPACK, RISCVBLAS). The tolerance level has now been set to 5×10^{-10} for both absolute and mantissa level checking. After various fixes, each of these paths now take a representative MATLAB script through translation, compilation with `riscv64-unknown-elf-gcc` 15.2.0, and execution on Spike without harness-level failure.

To measure the translated output, a `SPEED_*` directory family was added to the generated-code corpus that exercises the translated kernels at three problem sizes (`small` 2×2, `medium` 100×100, `big` 1000×1000) across nine operations: `addition`, `eigen`, `fourier`, `inverse`, `multiplication`, `power_complex`, `power_int`, `power_double`, and `transpose`. There also exist 61 other smaller test kernels such as `reindex`, `fourier`, and `filterM`. Each directory is built by a series of `make` commands, which regenerates the C source from the `.m` input, cross-compiles the binary, and records the Spike cycle count.

To measure the effect of this semester's translation-layer and library changes, the `make timediff` target replays every test through Spike and compares the retired-instruction count against a baseline captured at the 68-tests-passing checkpoint (before the current round of bug fixes began). The list of fixes discusses previously brought up the passing test count to 84/88. The 4 remaining comparison failures (`class_test`, `octaveIncludes`, `octaveTimer`, and `octave_syntax`) are due to pre-existing pipeline constraints. The Octave tests fail because they contain Octave-specific syntax configurations (such as `%source` directives and `endfunction`) which are not formally supported by the `tree-sitter-matlab` grammar, causing the pipeline to generate invalid C code. Similarly, `class_test` fails to generate evaluable C code because the transpiler currently lacks mappings for the `EEGLAB mmo` object structure. For `octaveTimer`, while the generated code successfully compiles and executes, the comparison fails due to system timer disparities; the C Standard Library's `time()` function provides 1-second resolution whereas MATLAB's `tic`/`toc` offers 1-microsecond precision, leading to a fundamental difference in timing.

In terms of cycle count improvements, Table 4.7 summarizes the results.

Table 4.7: TS-traversal end-to-end benchmark: retired instruction counts for tests that existed in the baseline (`compareall` at the 68-tests-passing checkpoint) compared against the current pipeline. Only representative rows are shown; the heavy `SPEED_big_*` workloads, the largest vectorization wins, the largest regressions, and newly added tests are included. Positive Δ indicates a speedup.

Test	Baseline (instret)	Current (instret)	Δ (%)
TODO: Heavy vectorized workloads (stable)			
SPEED_big_matrix_transpose	5.14×10^8	5.09×10^8	$+0.93$
SPEED_big_matrix_addition	7.08×10^8	7.08×10^8	$+0.00$
SPEED_big_matrix_fourier	9.06×10^8	9.06×10^8	$+0.00$
SPEED_big_matrix_multiplication	1.573×10^{10}	1.573×10^{10}	-0.00
SPEED_big_matrix_inverse	1.593×10^{10}	1.593×10^{10}	-0.00
SPEED_big_matrix_power_int	3.545×10^{10}	3.545×10^{10}	$+0.00$
SPEED_big_matrix_power_complex	7.422×10^{10}	7.422×10^{10}	$+0.00$
SPEED_big_matrix_power_double	7.422×10^{10}	7.422×10^{10}	$+0.00$
SPEED_big_matrix_eigen	1.117×10^{11}	1.117×10^{11}	-0.00
Largest vectorization wins			
SPEED_medium_matrix_transpose	1.78×10^7	1.3×10^7	$+26.75$
ceil_matrix	2.03×10^5	1.94×10^5	$+4.52$
round_matrix	2.09×10^5	2×10^5	$+4.33$
floor_matrix	2.04×10^5	1.96×10^5	$+4.05$
5 largest regressions			
fourier	2.93×10^7	5.75×10^7	-96.55
filterM	4.45×10^8	7.22×10^8	-62.38
elem_trig_matrix	1.41×10^6	2.1×10^6	-48.90
elem_scalarpower_matrix	2.32×10^6	3.26×10^6	-40.86
elem_power_matrix	2.09×10^6	2.8×10^6	-33.96

The observed speed improvement was less than initially expected due to the fact that ts-traversal changes and tests focused on improving and checking correctness and not performance. There was a hypothesis that the further vectorization of some functions like `transposeM` and `varM` would lead to improvements for their respective test cases. While most SPEED tests did not show improvement, `transposeM` did show notable improvement – 0.93% on `SPEED_big_transpose` and 26.75% on `SPEED_medium_transpose`. While the numbers seem tame, they tell a bigger story of improvement due to the `transposeM` calls in these test cases being made 1 and 100 times respectively. This suggests that with further vectorization, even if a heavy math function is called only a few times, having it vectorized enables quicker transition into desired idle states for patient safety. The main reason for the tests that regressed in their cycle counts is the print statement heavy nature of the tests. The definition of `print` was changed to print more precise values in order to have accurate tolerance checking and get more tests passing, causing significant slowdown for these tests. The lack of change in the `SPEED_big_fourier` test case supports the `print` statement hypothesis, as it has no `print` statements and thus was unaffected

by the change.

4.6.1 C-Octave Suite Residuals

There is a suite of Octave and C tests that mirror the same kernels tested in the MATLAB-vs-C comparison, only using Octave instead of MATLAB, which causes certain behavioral differences to be discussed in this section. The same tolerance gate (`tolerant_diff.py`, MANTISSA_TOL $=$ ABS_TOL $= 5 \times 10^{-10}$, EXP_JITTER $= 1$) is now wired into `tests_C_Octave/Makefile`'s `compare_outputs` target so the same numerical contract governs both surfaces. Following recent fixes to display handling, the number of passing tests went from 66 to 71/79. The 8 still-failing tests fail due to *FFT library divergence* or *error handling behavior differences*.

5 Related Work

The approach of this research is heavily informed by previous exploration into hardware-software co-design for implantable medical devices. The fundamental challenge of respecting the strict 1°C thermal limit has historically forced designers to rely on highly specialized ASICs. Kassiri et al.'s closed-loop neurostimulator [5], Neuralink's N1 ASIC [6], and the NeuroPace RNS System [7] represent the state-of-the-art implementations of this rigid baseline. These systems achieve unparalleled energy efficiency for their exact medical tasks by baking clinical logic into the physical silicon at the cost of algorithmic flexibility. Unlike these ASICs, HALO is built on the RVV open-source instruction set architecture (ISA). This enables its binaries to pick up improvements from every update of the open-source GCC backend without a tape-out. The Results chapter demonstrates the payoff: simply updating the compiler cut collective execution cost across the core arithmetic and logical operations (excluding Octave-mode comparison tests), and in three cases (`and` dbl×dbl, `and` cpx×dbl, `or` cpx×dbl) the auto-vectorized scalar code now beats the hand-written intrinsics outright.

5.1 Translation for High-Level Languages

Source-to-source code generation has been a topic of interest since the 1980s [11]. While this coincides with the release of MATLAB in the late 1970s, MATLAB did not have efficient transpilation tools until the 1990s with the introduction of FALCON by the University of Illinois at Urbana Champaign. DeRose et al. created a translator that was capable of compiling MATLAB into Fortran or C while giving developers the ability to "interactively apply optimizations and transformations to the code" [12]. FALCON followed a similar abstract syntax tree construction approach as this research does, but it suffered in the type inference phase. MathWorks have also been developing various MATLAB-to-C tools since the 1990s, such as `mcc` and `mex`, which were later replaced by MATLAB Coder [13]. MATLAB Coder is highly capable at parsing MATLAB scripts and generating optimized C and C++ code targeted primarily at advanced desktop processors and commercial ARM architectures. Similarly, for researchers utilizing Python–another dynamically typed language widely used in BCI data analysis and real-time processing pipelines–tools like Cython [14] offer similar benefits.

Although both MATLAB Coder and Cython are highly effective on standard architectures, they remain limited in use for this research as neither tool inherently supports the automated generation of targeted RISC-V Vector (RVV) intrinsics required to meet the ultra-low-power constraints of the HALO platform. Thus, the ts-traversal and RISCV-Matrix integration developed in this work under Professor Rajit Manohar's lab is neces-

sary to bridge the semantic gap between high-level BCI algorithm prototyping and the specific vector execution demands of the HALO embedded hardware.

6 Conclusions

The successful chronic implantation of next-gen BCIs is constrained by extreme power efficiency requirements due to patient safety concerns. This has historically been met through highly inflexible, monolithic ASIC hardware. HALO BCIs diverge from the norm by using various function-specific PEs and the RVV extension to introduce significant flexibility without compromising critical thermal limits. However, a highly optimized software compilation pipeline is required to make HALO and its clinical flexibility and capabilities accessible to the broader neuroscientific community. This project directly addresses this need by engineering an end-to-end toolchain that merges a TypeScript-based source-to-source compiler with a hardware-specific vectorization library, enabling the automatic transformation of high-level MATLAB algorithms into power-optimized RISC-V binaries.

Validation and benchmarking demonstrate that the vectorized matrix library successfully runs on Spike and yields near-consistent performance improvements. Analysis of the data has successfully isolated performance improvements driven by updated cross-compiler toolchains, while simultaneously identifying initial test-harness anomalies. The ts-traversal semantic translation layer has also been benchmarked and timed with successful results across various essential operations.

6.1 Future Work

The work presented here opens several concrete directions for future development, organized by component.

Continued vectorization. The RISCV-Matrix README flags dozens of kernels that remain unvectorized. The highest value vectorizations are likely trigonometric functions like `sinM` and the complex side of `argM` would provide significant speedups for BCI workloads due to the DSP-heavy nature of neural decoding. As discussed earlier, the vectorization of `carg` would speed-up the heavy Octave workloads from Table 4.1. However, this is likely a difficult endeavor as the lack of RVV trigonometric intrinsics require polynomial approximations to be built or pulled from a library like SLEEF [15]. Outside of these options, `popvarM` can be vectorized easily following the implementations of `varM`, `stdM`, `popstdM`. Finally, 1D non-double types can be vectorized for `varM`.

Integer overflow in `powM`. As noted in the C-Octave Suite Residuals discussion, `power_matrix` produces results exceeding `INT_MAX` $= 2^{31} - 1$ for `INTEGER` inputs. The Octave suite correctly computes these large values while the C suite gets stuck at `INT_MIN/INT_MAX`.

This warrants further investigation into `powM` and `floorM`, which both get called in the test suite.

LMUL audit. Several vectorized functions were flagged in the original README as having LMUL settings that "need checking." Using an LMUL that exceeds the number of live vector variables causes implicit register spills, negating the throughput benefit. An LMUL audit pass would identify and correct sub-optimal groupings.

Transpiler improvements. Support for MATLAB's `classdef` OOP is incomplete and would be a highest value change in order to enable supporting BCI workloads through EEGLAB: the `getClasses` pass discovers properties and methods but emits `initVar()` calls inside C `struct` definitions, which is illegal C. Separating struct declaration (header) from constructor initialization (function) would unblock the `class_test` suite.

HALO deployment. The hand-off to the BCI lab at the Yale School of Medicine will enable physical HALO testing under neuroscientist-written full workloads, where the 15 mW power envelope and 1 °C thermal limit will provide a real-world check on the cycle-count conclusions drawn here. Cache effects, TLB pressure, and memory bandwidth saturation – none of which are modeled by `rdinstret`-based benchmarking on Spike – will then become the dominant performance factors to drill into.

Bibliography

[1] Ioannis Karageorgos et al. "Hardware-Software Co-Design for Brain-Computer Interfaces". In: *2020 ACM/IEEE 47th Annual International Symposium on Computer Architecture (ISCA)*. 2020, pp. 391–404. DOI: 10.1109/ISCA45697.2020.00041.

[2] Karthik Sriram et al. "HALO: A Hardware–Software Co-Designed Processor for Brain–Computer Interfaces". In: *IEEE Micro* 43.3 (2023), pp. 64–72. DOI: 10.1109/MM.2023.3258907.

[3] Danielle Gruber. *ts-traversal: A Typescript Based MATLAB-to-C Compiler*. Version 1.0.0. 2024. URL: https://github.com/daniellegruber/ts-traversal.

[4] Zach Taylor and Xiayuan Wen. *RISCV-Matrix: A RISC-V Vectorized C Library for Matrix Manipulation*. Private internal repository, HALO Project. 2022.

[5] Hossein Kassiri et al. "Closed-Loop Neurostimulators: A Survey and A Seizure-Predicting Design Example for Intractable Epilepsy Treatment". In: *IEEE Transactions on Biomedical Circuits and Systems* 11.5 (2017), pp. 1026–1040. DOI: 10.1109/TBCAS.2017.2694638.

[6] Elon Musk and Neuralink. "An Integrated Brain-Machine Interface Platform With Thousands of Channels". In: *Journal of Medical Internet Research* 21.10 (2019), e16194. DOI: 10.2196/16194.

[7] Christianne N. Heck et al. "Two-year seizure reduction in adults with medically intractable partial onset epilepsy treated with responsive neurostimulation: Final results of the RNS System Pivotal trial". In: *Epilepsia* 55.3 (2014), pp. 432–441. DOI: 10.1111/epi.12534.

[8] Claudio Pipicelli. "Quantum Machine Learning: A Comparison Between Quantum and Classical Support Vector Machine". PhD thesis. Naples, Italy: Università degli Studi di Napoli Federico II, 2020. URL: http://www.fedoa.unina.it/13259/1/PhDThesisPipicelliFinal.pdf.

[9] Roger Ferrer Ibanez. URL: https://eupilot.eu/wp-content/uploads/2022/11/RISC-V-VectorExtension-1-1.pdf.

[10] GNU Octave Developers. *GNU Octave Manual: 8.4 Comparison Operators*. Accessed: 2026-04-20. 2025. URL: https://docs.octave.org/latest/Comparison-Ops.html (visited on 04/20/2026).

[11] Roger Taylor and Phil Lemmons. "Upward Migration—Part 1: Translators". In: *BYTE Magazine* 7.6 (June 1982), pp. 321–344. URL: https://web.archive.org/web/20200116024623/https://tech-insider.org/personal-computers/research/acrobat/8206-b.pdf.

[12] Luiz De Rose et al. “FALCON: A MATLAB Interactive Restructuring Compiler”. In: *Languages and Compilers for Parallel Computing (LCPC’95)*. Vol. 1033. Lecture Notes in Computer Science. Springer, 1996, pp. 269–288. DOI: 10.1007/bfb0014201.

[13] MathWorks. *MATLAB Coder: Generate C and C++ code from MATLAB code*. 2024. URL: https://www.mathworks.com/products/matlab-coder.html.

[14] Stefan Behnel et al. “Cython: The Best of Both Worlds”. In: *Computing in Science & Engineering* 13.2 (2011), pp. 31–39.

[15] Naoki Shibata and contributors. *SLEEF: SIMD Library for Evaluating Elementary Functions*. Accessed: 2026-04-27. URL: https://sleef.org.

A Full Binary Benchmark Data

A.1 Full Binary Ratio Comparison

Table A.1 presents the complete vec/ref ratio comparison for all nine type combinations across all binary operations.

Table A.1: Vec/Ref ratios at $N \approx 1{,}050{,}625$ — all type combos for binary operations.

Test	Type Combo	Grace	Current	Δ
and	i×i	0.63752	0.63143	-0.00609
	i×d	0.73894	0.73078	-0.00816
	i×c	0.29241	0.30427	+0.01186
	d×i	0.51947	0.51219	-0.00728
	d×d	0.49314	2.67753	+2.18439
	d×c	0.43051	0.44859	+0.01808
	c×i	0.42433	0.47004	+0.04571
	c×d	0.40785	2.46448	+2.05663
	c×c	0.36534	0.41729	+0.05195
divide	i×i	0.28021	0.82317	+0.54296
	i×d	0.30110	0.29819	-0.00291
	i×c	0.21632	0.10596	-0.11036
	d×i	0.30110	0.29819	-0.00291
	d×d	0.32490	0.32176	-0.00314
	d×c	0.22214	0.10690	-0.11524
	c×i	0.21408	0.12221	-0.09187
	c×d	0.21987	0.12398	-0.09589
	c×c	0.23512	0.13319	-0.10193
equal	i×i	0.17294	0.13845	-0.03449
	i×d	0.18083	0.14332	-0.03751
	i×c	0.16096	0.12921	-0.03175
	d×i	0.18083	0.14332	-0.03751
	d×d	0.19126	0.15016	-0.04110
	d×c	0.15742	0.12757	-0.02985
	c×i	0.16096	0.12921	-0.03175
	c×d	0.15742	0.12757	-0.02985
	c×c	0.16057	0.12889	-0.03168

Test	Type Combo	Grace	Current	Δ
isEqual	i×i	0.67105	0.65753	-0.01352
	i×d	0.82667	0.75000	-0.07667
	i×c	1.11940	1.10000	-0.01940
	d×i	0.83784	0.75000	-0.08784
	d×d	0.79710	0.73134	-0.06576
	d×c	1.11111	1.08772	-0.02339
	c×i	1.13636	1.11667	-0.01969
	c×d	0.01514	0.00899	-0.00615
	c×c	1.17188	1.12069	-0.05119
matlab_ge	i×i	0.28173	0.27919	-0.00254
	i×d	0.25460	0.25105	-0.00355
	i×c	0.25526	0.25171	-0.00355
	d×i	0.23476	0.23148	-0.00328
	d×d	0.25328	0.82751	+0.57423
	d×c	0.25460	0.25171	-0.00289
	c×i	0.23537	0.23209	-0.00328
	c×d	0.25460	0.25171	-0.00289
	c×c	0.25394	0.25105	-0.00289
matlab_gt	i×i	0.28173	0.27919	-0.00254
	i×d	0.25460	0.25105	-0.00355
	i×c	0.25526	0.25171	-0.00355
	d×i	0.23476	0.23148	-0.00328
	d×d	0.25328	0.82751	+0.57423
	d×c	0.27810	0.27495	-0.00315
	c×i	0.23537	0.23209	-0.00328
	c×d	0.27810	0.27495	-0.00315
	c×c	0.25394	0.25105	-0.00289
matlab_le	i×i	0.28173	0.27919	-0.00254
	i×d	0.23476	0.23148	-0.00328
	i×c	0.23537	0.23209	-0.00328
	d×i	0.25460	0.25105	-0.00355
	d×d	0.27666	0.82751	+0.55085
	d×c	0.25460	0.25171	-0.00289
	c×i	0.25526	0.25171	-0.00355
	c×d	0.25460	0.25171	-0.00289
	c×c	0.27738	0.27423	-0.00315
matlab_lt	i×i	0.28173	0.27919	-0.00254
	i×d	0.23476	0.23148	-0.00328
	i×c	0.23537	0.23209	-0.00328
	d×i	0.25460	0.25105	-0.00355
	d×d	0.27666	0.82751	+0.55085

Test	Type Combo	Grace	Current	Δ
	d×c	0.27810	0.27495	-0.00315
	c×i	0.25526	0.25171	-0.00355
	c×d	0.27810	0.27495	-0.00315
	c×c	0.27738	0.27423	-0.00315
minus	i×i	0.76190	0.75801	-0.00389
	i×d	0.30110	0.29819	-0.00291
	i×c	0.39134	0.38842	-0.00292
	d×i	0.30110	0.29819	-0.00291
	d×d	0.32490	0.32176	-0.00314
	d×c	0.41179	0.40830	-0.00349
	c×i	0.41179	0.40873	-0.00306
	c×d	0.43465	0.43143	-0.00322
	c×c	0.39210	0.48982	+0.09772
mtimes	i×i	0.05943	0.04814	-0.01129
	i×d	0.05590	0.04422	-0.01168
	i×c	0.04720	0.03477	-0.01243
	d×i	0.05590	0.04422	-0.01168
	d×d	0.05213	0.04010	-0.01203
	d×c	0.04721	0.03469	-0.01252
	c×i	0.04720	0.03477	-0.01243
	c×d	0.04721	0.03469	-0.01252
	c×c	0.04433	0.03176	-0.01257
ne	i×i	0.16324	0.13211	-0.03113
	i×d	0.17069	0.13676	-0.03393
	i×c	0.15324	0.12401	-0.02923
	d×i	0.17069	0.13676	-0.03393
	d×d	0.17990	0.14295	-0.03695
	d×c	0.14986	0.12244	-0.02742
	c×i	0.15324	0.12401	-0.02923
	c×d	0.14986	0.12244	-0.02742
	c×c	0.15286	0.12370	-0.02916
octave_ge	i×i	0.28173	0.27919	-0.00254
	i×d	0.25460	0.25105	-0.00355
	i×c	0.14918	0.13115	-0.01803
	d×i	0.23476	0.23148	-0.00328
	d×d	0.25328	0.82751	+0.57423
	d×c	0.15009	0.13215	-0.01794
	c×i	0.14991	0.13262	-0.01729
	c×d	0.15097	0.13389	-0.01708
	c×c	0.14158	0.12297	-0.01861
octave_gt	i×i	0.28173	0.27919	-0.00254

Test	Type Combo	Grace	Current	Δ
	i×d	0.25460	0.25105	-0.00355
	i×c	0.14918	0.13115	-0.01803
	d×i	0.23476	0.23148	-0.00328
	d×d	0.25328	0.82751	+0.57423
	d×c	0.15009	0.13215	-0.01794
	c×i	0.14991	0.13262	-0.01729
	c×d	0.15097	0.13389	-0.01708
	c×c	0.14158	0.12297	-0.01861
octave_le	i×i	0.28173	0.27919	-0.00254
	i×d	0.23476	0.23148	-0.00328
	i×c	0.15024	0.13262	-0.01762
	d×i	0.25460	0.25105	-0.00355
	d×d	0.27666	0.82751	+0.55085
	d×c	0.15117	0.13365	-0.01752
	c×i	0.14886	0.13115	-0.01771
	c×d	0.14990	0.13261	-0.01729
	c×c	0.14103	0.12221	-0.01882
octave_lt	i×i	0.28173	0.27919	-0.00254
	i×d	0.23476	0.23148	-0.00328
	i×c	0.15024	0.13262	-0.01762
	d×i	0.25460	0.25105	-0.00355
	d×d	0.27666	0.82751	+0.55085
	d×c	0.15117	0.13365	-0.01752
	c×i	0.14886	0.13115	-0.01771
	c×d	0.14990	0.13261	-0.01729
	c×c	0.14103	0.12221	-0.01882
or	i×i	0.64781	0.64162	-0.00619
	i×d	0.78077	0.77214	-0.00863
	i×c	0.34462	0.33687	-0.00775
	d×i	0.52858	0.52118	-0.00740
	d×d	0.51821	0.51066	-0.00755
	d×c	0.49995	0.49200	-0.00795
	c×i	0.48494	0.47004	-0.01490
	c×d	0.47687	2.39202	+1.91515
	c×c	0.46204	0.44753	-0.01451
pairwise_max	i×i	0.00111	0.00083	-0.00028
	i×d	0.00363	0.00287	-0.00076
	i×c	0.13935	0.12188	-0.01747
	d×i	0.00363	0.00287	-0.00076
	d×d	0.00404	0.00406	+0.00002
	d×c	0.14133	0.12386	-0.01747

Test	Type Combo	Grace	Current	Δ
	c×i	0.14012	0.12332	-0.01680
	c×d	0.14224	0.12551	-0.01673
	c×c	0.13472	0.11656	-0.01816
pairwise_min	i×i	0.00111	0.00083	-0.00028
	i×d	0.00363	0.00287	-0.00076
	i×c	0.14043	0.12332	-0.01711
	d×i	0.00363	0.00287	-0.00076
	d×d	0.00404	0.00406	+0.00002
	d×c	0.14244	0.12534	-0.01710
	c×i	0.13904	0.12188	-0.01716
	c×d	0.14113	0.12423	-0.01690
	c×c	0.13414	0.11580	-0.01834
plus	i×i	0.76190	0.75801	-0.00389
	i×d	0.30110	0.29819	-0.00291
	i×c	0.39053	0.38761	-0.00292
	d×i	0.30110	0.29819	-0.00291
	d×d	0.32490	0.32176	-0.00314
	d×c	0.41094	0.40787	-0.00307
	c×i	0.39049	0.38757	-0.00292
	c×d	0.41094	0.40787	-0.00307
	c×c	0.39210	0.48982	+0.09772
scale	i×i	0.80681	0.80392	-0.00289
	i×d	0.32490	0.32243	-0.00247
	i×c	0.41012	0.45743	+0.04731
	d×i	0.38970	0.38678	-0.00292
	d×d	0.35316	0.35048	-0.00268
	d×c	0.43285	0.48625	+0.05340
	c×i	0.46127	0.56105	+0.09978
	c×d	0.43465	0.48778	+0.05313
	c×c	0.31024	0.30722	-0.00302
times	i×i	0.76190	0.75801	-0.00389
	i×d	0.30110	0.29819	-0.00291
	i×c	0.29864	0.29604	-0.00260
	d×i	0.30110	0.29819	-0.00291
	d×d	0.32490	0.32176	-0.00314
	d×c	0.31024	0.30754	-0.00270
	c×i	0.29861	0.29601	-0.00260
	c×d	0.31024	0.30754	-0.00270
	c×c	0.30169	0.29725	-0.00444

A.2 Binary Operation Cycle Counts

Table A.2 presents the raw vectorized and reference cycle counts for the diagonal type combinations (int×int, double×double, complex×complex) of all binary operations.

Table A.2: Vectorized and reference cycle counts at $N \approx 1{,}050{,}625$ — diagonal type combos for binary operations.

Test	Type	Grace vec	Cur. vec	Grace ref	Cur. ref
and	i×i	7.0M	7.0M	11.0M	11.0M
	d×d	5.4M	5.3M	10.9M	2.0M
	c×c	5.5M	5.4M	15.1M	13.0M
divide	i×i	4.0M	4.0M	14.4M	4.8M
	d×d	4.0M	3.9M	12.3M	12.2M
	c×c	8.5M	8.2M	36.1M	61.3M
equal	i×i	3.1M	3.0M	17.7M	21.9M
	d×d	3.2M	3.1M	16.6M	20.8M
	c×c	3.3M	3.2M	20.8M	25.0M
isEqual	i×i	51	48	76	73
	d×d	55	49	69	67
	c×c	75	65	64	58
matlab_ge	i×i	3.1M	3.0M	10.9M	10.9M
	d×d	3.1M	3.1M	12.4M	3.8M
	c×c	3.2M	3.1M	12.4M	12.4M
matlab_gt	i×i	3.1M	3.0M	10.9M	10.9M
	d×d	3.1M	3.1M	12.4M	3.8M
	c×c	3.2M	3.1M	12.4M	12.4M
matlab_le	i×i	3.1M	3.0M	10.9M	10.9M
	d×d	3.1M	3.1M	11.4M	3.8M
	c×c	3.2M	3.1M	11.4M	11.4M
matlab_lt	i×i	3.1M	3.0M	10.9M	10.9M
	d×d	3.1M	3.1M	11.4M	3.8M
	c×c	3.2M	3.1M	11.4M	11.4M
minus	i×i	13.5M	13.4M	17.7M	17.7M
	d×d	4.0M	3.9M	12.3M	12.2M
	c×c	8.0M	7.9M	20.3M	16.1M
mtimes	i×i	925.2K	740.7K	15.6M	15.4M
	d×d	794.5K	611.2K	15.2M	15.2M
	c×c	1.8M	1.3M	41.1M	41.1M
ne	i×i	3.1M	3.0M	18.7M	22.9M

Test	Type	Grace vec	Cur. vec	Grace ref	Cur. ref
	d×d	3.2M	3.1M	17.7M	21.9M
	c×c	3.3M	3.2M	21.9M	26.1M
octave_ge	i×i	3.1M	3.0M	10.9M	10.9M
	d×d	3.1M	3.1M	12.4M	3.8M
	c×c	303.5M	270.2M	2143.5M	2197.5M
octave_gt	i×i	3.1M	3.0M	10.9M	10.9M
	d×d	3.1M	3.1M	12.4M	3.8M
	c×c	303.5M	270.2M	2143.5M	2197.5M
octave_le	i×i	3.1M	3.0M	10.9M	10.9M
	d×d	3.1M	3.1M	11.4M	3.8M
	c×c	303.5M	270.2M	2151.9M	2211.2M
octave_lt	i×i	3.1M	3.0M	10.9M	10.9M
	d×d	3.1M	3.1M	11.4M	3.8M
	c×c	303.5M	270.2M	2151.9M	2211.2M
or	i×i	7.0M	7.0M	10.9M	10.9M
	d×d	5.4M	5.3M	10.3M	10.3M
	c×c	5.5M	5.4M	11.9M	12.1M
pairwise_max	i×i	66.4K	49.7K	60.0M	59.8M
	d×d	132.0K	99.0K	32.7M	24.4M
	c×c	293.1M	259.9M	2175.6M	2229.4M
pairwise_min	i×i	66.4K	49.7K	60.0M	59.8M
	d×d	132.0K	99.0K	32.7M	24.4M
	c×c	293.1M	259.9M	2185.1M	2244.1M
plus	i×i	13.5M	13.4M	17.7M	17.7M
	d×d	4.0M	3.9M	12.3M	12.2M
	c×c	8.0M	7.9M	20.3M	16.1M
scale	i×i	13.4M	13.4M	16.6M	16.6M
	d×d	4.0M	3.9M	11.2M	11.2M
	c×c	7.9M	7.8M	25.6M	25.5M
times	i×i	13.5M	13.4M	17.7M	17.7M
	d×d	4.0M	3.9M	12.3M	12.2M
	c×c	8.0M	7.9M	26.6M	26.6M